My name is Moaaz Tameer Islam. I'm completing a Master's in Data Science at Uppsala University, graduating June 2026, and can start August 2026.

Growing up in a compound for doctors due to my dad's job, and with friends studying medicine, I developed an interest in healthcare early on. The first time I worked on the healthtech intersection was after studying ML in my bachelor's. My mind went to how biases in healthcare literature would be cemented in the upcoming age of AI healthcare. I compiled my findings into a blog and messaged researchers working on such models at the time, proposing methods to check fairness. Received a few positive responses, but a lack of resources prevented any actual work. That's what motivated me to pursue a master's and be part of an institute again.

Satisfied with my ML knowledge base, I focused on applying it. I took a health innovations course covering the EU healthtech landscape and the end-to-end process of bringing innovations through it. A hospital project course investigating EHR interoperability challenges in Sweden, learning the gap between implemented and existing standards. A small consultancy project forecasting patient arrivals at an emergency department. Most recently, I've started a pilot with a hospital in Saudi Arabia on using AI to read electrophoresis results and speed up lab turnaround times. None are research-heavy projects individually, but the breadth of exposure means I understand the structures and common problems. A PhD coming from that awareness is more likely to produce something workers actually want to use, and that fits the system.

Part of why I want a PhD, and why I left a data analyst role at Samsung despite it being a good position on paper, is that I need room to actually explore a problem. The role was in the innovation department but turned out to be fairly constrained, down to not being allowed to install Python on a work laptop. I don't enjoy the repetitive product analytics side of industry data work, and I found that out clearly enough. What appeals to me about a PhD is the open-endedness: not knowing how an experiment will turn out, having the time to follow a problem properly, and working across the range of stakeholders who matter to the outcome. Whether I stay in academia after is something I'm genuinely undecided on, but four years in the system will clarify that.

The specific research question here is one I find compelling beyond its technical difficulty. A friend of mine received a false negative on cancer spread, and the consequences of missing metastasis, even briefly, are serious and fast-moving. The promise of ctDNA as a signal is that it offers a less invasive and more continuous window into what's happening than imaging alone. Accurate ctDNA interpretation also opens a new data point that new research can build on, which is the kind of foundational work that compounds over time. Beyond the human cost, imaging is expensive, and healthcare budgets are under increasing pressure. A reliable blood-based signal can reduce unnecessary scans and catch what scans miss.

The technical work I'm most proud of in my master's is the multimodal pipeline I built for my thesis. I was classifying harmful comments on short-form video platforms, and a portion of the data was image and GIF-based. The pipeline ran a vision-language model to generate descriptions, then fed those into the text classifier. Because I was doing everything alone, from collecting, annotating, to building, I had a much better feel for what would work. I essentially thought, “What information helped me decide if something was harmful?”. So I came up with annotater notes, describing GIFs the same way I describe Gen Z slang to my parents, and it worked great for the accuracy. Similarly, I’d come across comments that, just by their content, could be harmless or harmful. But given the vibe of the entire comment section, it became clear what their intention is, so I included multiple passes in one of the architectures. I finally felt confident in making design choices and knew what was possible and how to implement it, compared to just running the latest models on the data.

I don't have a computational biology background. That's a gap I'd expect to close through coursework and the collaboration with Karolinska, and it's the kind of domain knowledge that's acquirable given the right environment. My academic interests have consistently landed at the point where a hard ML problem has a concrete real-world use case that depends on understanding the feasibility and picking the right option, ideally, healthcare-related. This project has those properties. I'd rather spend four years on something where the answer matters than on something technically clean but disconnected from anything that changes how a decision gets made.